

obstacles avoiding robots

```
#include <Servo.h>
```

```
int trigPin = 9;
```

```
int echoPin = 10;
```

```
int IN1 = 2;
```

```
int IN2 = 3;
```

```
int IN3 = 4;
```

```
int IN4 = 7;
```

```
int servoPin = 11;
```

```
Servo myServo;
```

```
long duration;
```

```
int distance;
```

```
void setup()
```

```
{
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
  pinMode(IN1, OUTPUT);
```

```
  pinMode(IN2, OUTPUT);
```

```
  pinMode(IN3, OUTPUT);
```

```
  pinMode(IN4, OUTPUT);
```

```
  myServo.attach(servoPin);
```

```
  myServo.write(90);
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop()
```

```
{
```

```
  distance = getDistance();
```

```
  Serial.print("Distance: ");
```

```
  Serial.print(distance);
```

```
  Serial.println(" cm");
```

```
  if (distance > 20)
```

```

{
    forward();
}
else
{
    stopRobot();
    delay(300);

    backward();
    delay(400);

    stopRobot();
    delay(300);

    // Look left
    myServo.write(150);
    delay(500);
    int leftDistance = getDistance();

    // Look right
    myServo.write(30);
    delay(500);
    int rightDistance = getDistance();

    // Return to center
    myServo.write(90);
    delay(300);

    if (leftDistance > rightDistance)
    {
        turnLeft();
        delay(600);
    }
    else
    {
        turnRight();
        delay(600);
    }

    stopRobot();
    delay(200);
}
}

```

```

int getDistance()

```

```
{  
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
  
    digitalWrite(trigPin, LOW);  
  
    duration = pulseIn(echoPin, HIGH, 30000);  
  
    if (duration == 0)  
    {  
        return 400;  
    }  
  
    return duration * 0.034 / 2;  
}
```

```
void forward()  
{  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}
```

```
void backward()  
{  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}
```

```
void turnLeft()  
{  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}
```

```
void turnRight()
{
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);

    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
}
```

```
void stopRobot()
{
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);

    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
}
```